Sungjae Choi

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Research Interests

3D computer vision with a focus on 3D Gaussian splatting and multi-view geometry foundation models: radiance-field segmentation, few-shot reconstruction, and training-free scaling of feed-forward geometry transformers to long sequences. Emerging interests in dynamic scene reconstruction and physics-consistent 4D representations.

KAIST *Daejeon, Korea*
M.S. in Electrical Engineering, SIIT Lab (Advisor: Prof. Junmo Kim) Mar 2025 – Feb 2027 (expected)

GPA: 3.78 / 4.30 **KAIST** *Daejeon, Korea*
B.S. in Electrical Engineering, Minor in Cultural Technology Mar 2020 – Feb 2025

GPA: 3.92 / 4.30

Publications

First-author papers at international conferences

- **Sungjae Choi**, Hanna Bae, Sunghyun Baek, Junmo Kim. *VGGT-Bridge: Beyond Sequential Pose Graphs via Coarse-Stride Skip Edges.* Asian Conference on Computer Vision (**ACCV**), 2026.
- **Sungjae Choi**, Seunghee Koh, Junmo Kim. *PePESeg3D: Perception Prior Enhances Multi-Scale Segmentation for 3D Gaussian Splatting.* British Machine Vision Conference (**BMVC**), 2026. Earlier version presented at the CVPR 2026 Workshop on 3D Multi-View (3DMV).

Co-authored

- Seunghee Koh, **Sungjae Choi**, Minchan Kwon, Sunghyun Baek, Junmo Kim. *</think> Doesn't Stop Reasoning: Analysis of Spurious CoT Termination.* Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2026 (Main). arXiv:2609.03633

Under review

- **Sungjae Choi**, et al. *VistaGS: Multi-View Geometry Foundation Priors for View-Consistent Few-Shot 3D Gaussian Splatting.* Under review.

Domestic publications (undergraduate research)

- Kyoungwhan Mheen, Heejeong Ko, **Sungjae Choi**, Sangho Yoon. *Worki-Yo: AR-Driven Stretching in Office System for Workplace Health.* HCI Korea, 2025.
- Heejeong Ko, Kyoungwhan Mheen, **Sungjae Choi**, Sangho Yoon, Woontack Woo. *HMD-Based Trajectory Extraction for Seated Arm Stretching Based on Linear and Angular Trajectory Classification.* KIISE Transactions on Computing Practices, 32(9):440–445, 2026. Extended from the Korea Software Congress (KSC) 2024 paper, **Best Presentation Paper Award**.

Patent

- Gyuseop Song, Wonjun Lee, Jaechul Kim, Byungji Park, **Sungjae Choi**, Dongho Lee, Hyunse Suh, Dongsu Kim. *Stone Size Estimation Method.* KR Patent Application 10-2023-0031846, 2023.

Experience

Hyundai Motor Company *Seoul, Korea*
Research Intern (EE Co-op), Autonomous Driving Perception Development Team Sep 2023 – Feb 2024

- Built a 3D verification tool for monocular distance estimation used in perception validation.

ROEN Surgical *Daejeon, Korea*
Research Intern (EE Externship) Jul 2022 – Aug 2022

- Developed kidney stone detection and size estimation for surgical imaging; co-inventor on the resulting patent.

Honors and Awards

- Best Presentation Paper Award, Korea Software Congress (KSC), 2024
- Encouragement Award, KAIST Failure Institute Essay Contest

Service and Leadership

- Manager of GPU server infrastructure, SIIT Lab, KAIST (2025 – present)
- Student Leader (Head of Delegation), 19th KAIST–POSTECH Science War

Skills

Programming: Python, PyTorch, C/C++
3D Vision: 3D Gaussian Splatting, NeRF, VGGT-family geometry transformers, SLAM/pose graph optimization
Systems: Linux server administration
Languages: Korean (native), English (fluent)